



Lesson 4.3 — Horizontal and Slant Asymptotes

Big idea: Horizontal and slant asymptotes describe what the graph does as $x \rightarrow \pm\infty$. They're decided by the degrees of the numerator and denominator.

Quick review

Compare degrees of numerator (n) and denominator (d):

- $n < d \rightarrow$ horizontal asymptote $y = 0$.
- $n = d \rightarrow$ horizontal asymptote $y = (\text{leading coeff of } P) / (\text{leading coeff of } Q)$.
- $n = d + 1 \rightarrow$ slant (oblique) asymptote, found by polynomial division (use the quotient, ignore remainder).
- $n > d + 1 \rightarrow$ no horizontal or slant; end behavior follows a higher-degree polynomial.

Crossing: a graph CAN cross its horizontal asymptote at finite x (asymptotes only describe the ends).

Worked example

Worked example. Find the asymptotes of $f(x) = (x^2 + 1) / (x - 2)$.

Degrees: 2 and 1, so $n = d + 1 \rightarrow$ slant asymptote. Divide: $(x^2 + 1)/(x - 2) = x + 2 + 5/(x - 2)$. Slant: $y = x + 2$. Vertical asymptote: $x = 2$.

Warm-ups

1. Find the horizontal asymptote of $f(x) = 5 / (x^2 + 1)$.
2. Find the horizontal asymptote of $f(x) = (3x + 1)/(x - 2)$.
3. Does $f(x) = (x^3 + 1)/(x - 1)$ have a horizontal asymptote?

Practice

1. Find horizontal/slant asymptote(s) of $f(x) = (2x^2 + 3)/(x^2 + x - 6)$.
2. Find horizontal/slant asymptote(s) of $f(x) = (x^2 - 4x)/(x - 1)$.
3. Does the graph of $f(x) = (x - 1)/(x^2 + 1)$ cross its horizontal asymptote? Find any crossing points.

Challenge

1. Build a rational function with vertical asymptote $x = 2$, slant asymptote $y = x + 1$, and y -intercept 0.
2. $f(x) = (ax^2 + bx + c)/(x^2 + 1)$. For what conditions on a , b , c does f have horizontal asymptote $y = 3$ and pass through $(0, 5)$?



Answer key — Lesson 4.3

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $y = 0$ (since $n < d$).
2. $y = 3$ (ratio of leading coefficients, $n = d$).
3. No — $n > d + 1$ (3 vs 1), so no horizontal/slant. End behavior follows x^2 .

Practice

1. $n = d$, ratio of leading coefficients $2/1 \rightarrow y = 2$.
2. $n = d + 1$; divide $(x^2 - 4x)$ by $(x - 1)$: $x - 3$ with remainder -3 . Slant: $y = x - 3$. (Also VA at $x = 1$.)
3. HA: $y = 0$ ($n < d$). Crosses where $(x - 1)/(x^2 + 1) = 0 \Rightarrow x = 1$. Point: $(1, 0)$.

Challenge

1. $f(x) = (x^2 + x + k)/(x - 2)$. Slant from division: $x + 3$ (need this = $x + 1$ means quotient is $x + 1$) — recompute: divide $(x^2 + x + k)$ by $(x - 2)$: $x + 3$ with remainder $k + 6$. So slant is $y = x + 3$, not $x + 1$. To get slant $y = x + 1$, choose numerator $(x^2 - x + k)$: divide $\rightarrow x + 1$ with remainder $k + 2$. Set $f(0) = k/(-2) = 0 \Rightarrow k = 0$. So $f(x) = (x^2 - x)/(x - 2) = x(x - 1)/(x - 2)$. Check: VA $x = 2$ ✓, slant $y = x + 1$ ✓, $f(0) = 0$ ✓.
2. $n = d$, HA = $a/1 = 3 \Rightarrow a = 3$. $f(0) = c/1 = 5 \Rightarrow c = 5$. b is free.